

OLTP and OLAP

DATABASE DESIGN



Lis Sulmont
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Our motivating question:

How should we organize and manage data?

- **Schemas:** *How should my data be logically organized?*
- **Normalization:** *Should my data have minimal dependency and redundancy?*
- **Views:** *What joins will be done most often?*
- **Access control:** *Should all users of the data have the same level of access*
- **DBMS:** *How do I pick between all the SQL and noSQL options?*
- and more!

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It depends on the intended use of the data.

Approaches to processing data

OLTP

Online Transaction Processing



OLAP

Online Analytical Processing



Some concrete examples

OLTP tasks

- Find the price of a book
- Update latest customer transaction
- Keep track of employee hours

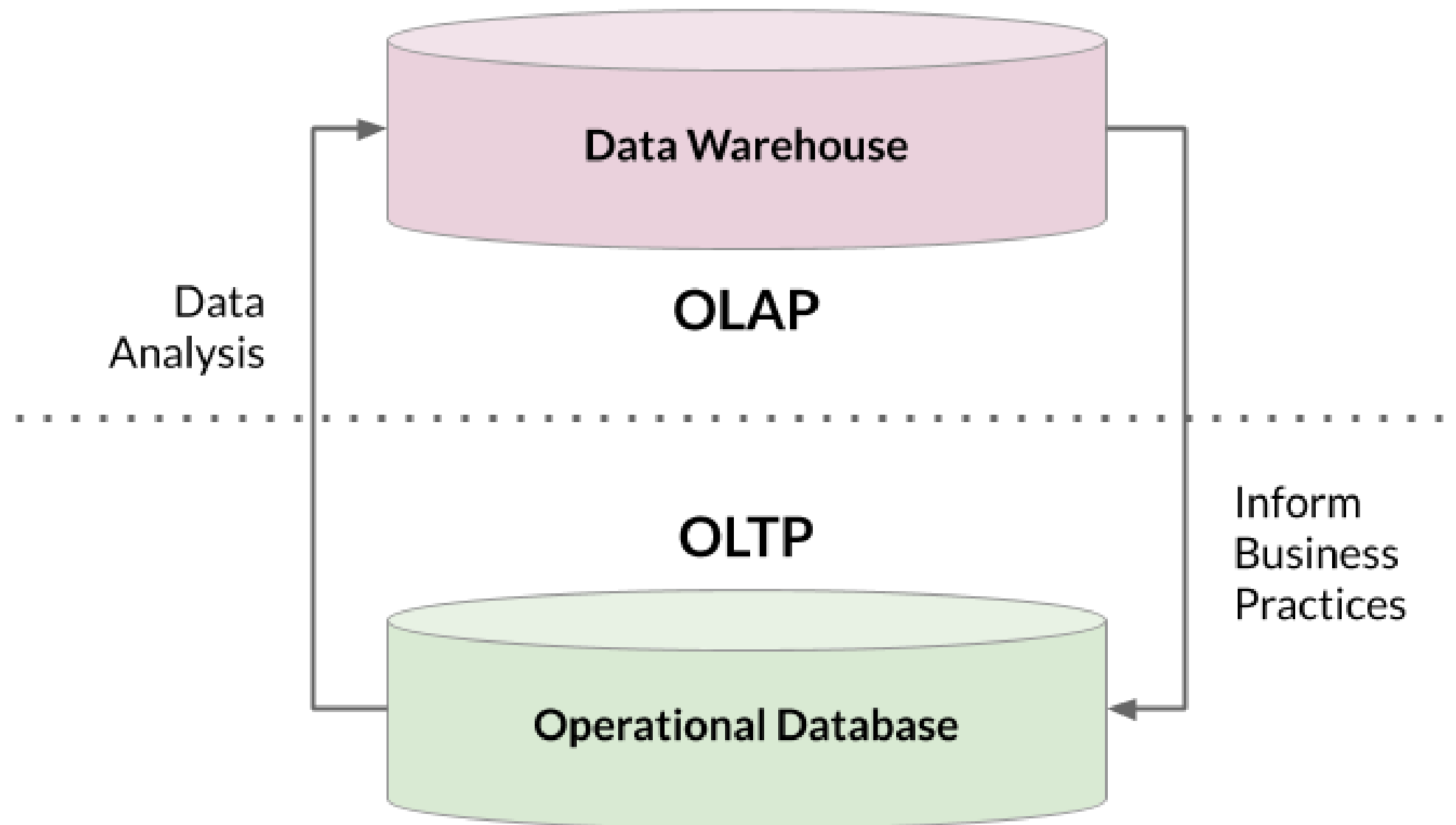
OLAP tasks

- Calculate books with best profit margin
- Find most loyal customers
- Decide employee of the month

OLAP vs. OLTP

	OLTP	OLAP
<i>Purpose</i>	support daily transactions	report and analyze data
<i>Design</i>	application-oriented	subject-oriented
<i>Data</i>	up-to-date, operational	consolidated, historical
<i>Size</i>	snapshot, gigabytes	archive, terabytes
<i>Queries</i>	simple transactions & frequent updates	complex, aggregate queries & limited updates
<i>Users</i>	thousands	hundreds

Working together



Takeaways

- Step back and figure out business requirements
- Difference between OLAP and OLTP
- OLAP? OLTP? Or something else?

Let's practice!
DATABASE DESIGN

Storing data

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Structuring data

1. Structured data

- Follows a schema
 - Defined data types & relationships
- _e.g., SQL, tables in a relational database _*

3. Semi-structured data

- Does not follow larger schema
 - Self-describing structure
- e.g., NoSQL, XML, JSON*

2. Unstructured data

- Schemaless
 - Makes up most of data in the world
- e.g., photos, chat logs, MP3*

Example of a JSON file

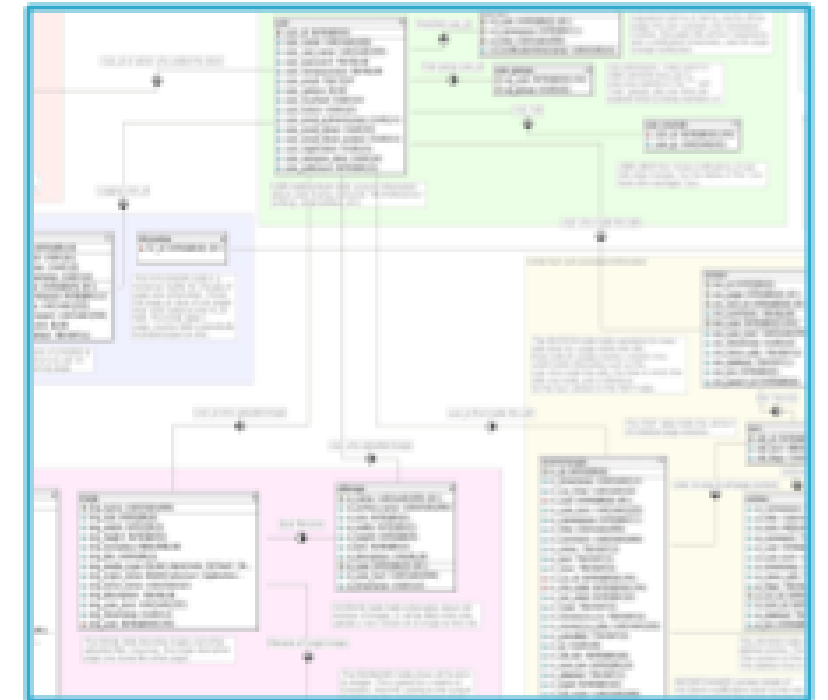
```
"user": {  
    "profile_use_background_image": true,  
    "statuses_count": 31,  
    "profile_background_color": "CODEED",  
    "followers_count": 3066,  
    ...  
}
```

Structuring data

Easier to Analyze



```
<?xml version="1.0"
  encoding="iso-8859-1" ?>
<languages>
  <language id="fr">
    <name lang="fr">Français</name>
    <name lang="en">French</name>
    <name lang="es">Frances</name>
    <name lang="de">Französisch</name>
    <name lang="eo">Franca</name>
  </language>
</languages>
```



More Flexibility and Scalability

¹ Flower by Sam Oth and Database Diagram by Nick Jenkins via Wikimedia Commons
https://commons.wikimedia.org/wiki/File/Languages_xml.png

Storing data beyond traditional databases

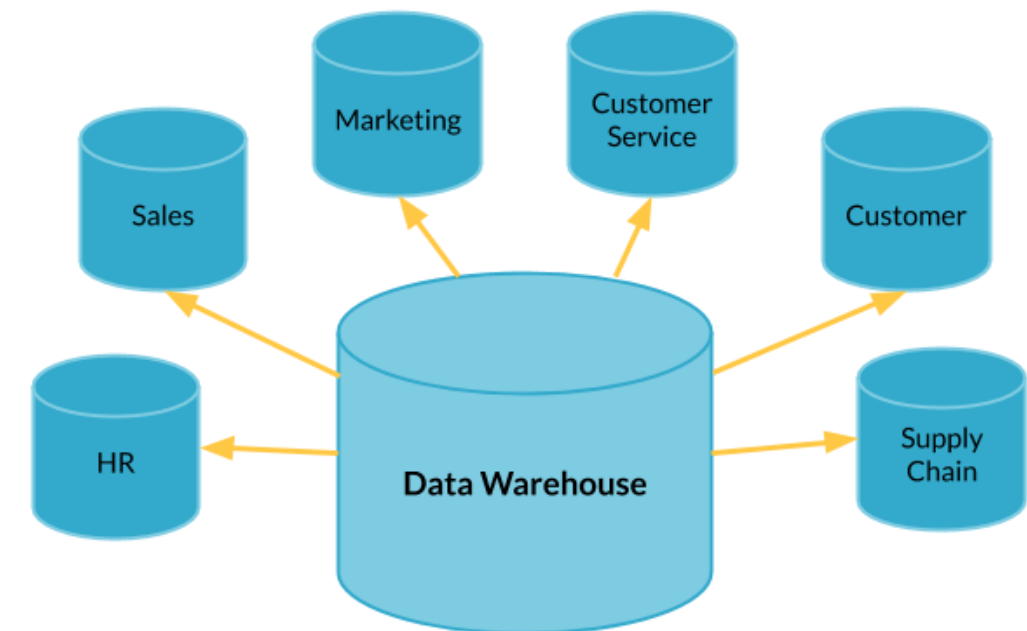
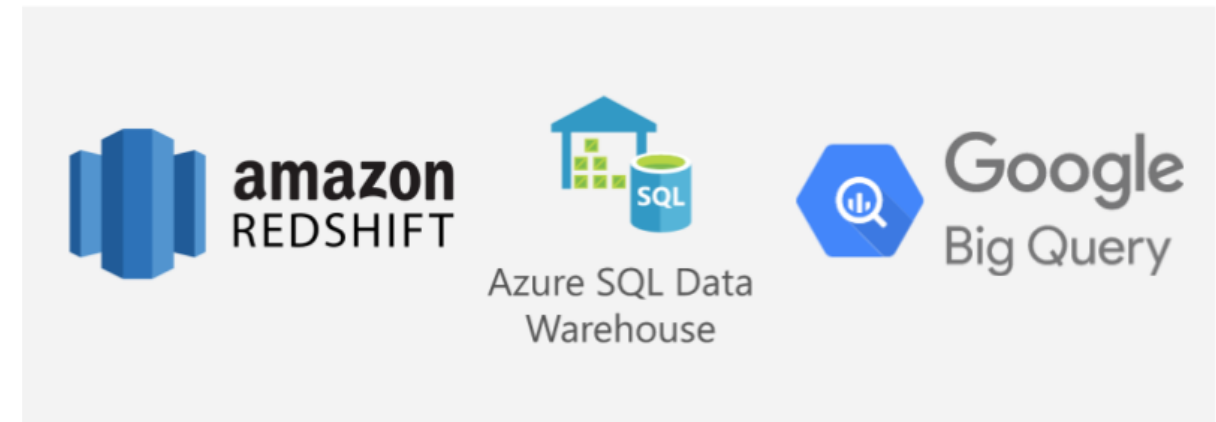
- **Traditional databases**
 - For storing real-time relational structured data → OLTP
- **Data warehouses**
 - For analyzing archived structured data → OLAP
- **Data lakes**
 - For storing data of all structures = flexibility and scalability
 - For analyzing **big data**

Data warehouses

- Optimized for analytics - OLAP
 - Organized for reading/aggregating data
 - Usually read-only
- Contains data from multiple sources
- Massively Parallel Processing (MPP)
- Typically uses a denormalized schema and dimensional modeling

Data marts

- Subset of data warehouses
- Dedicated to a specific topic

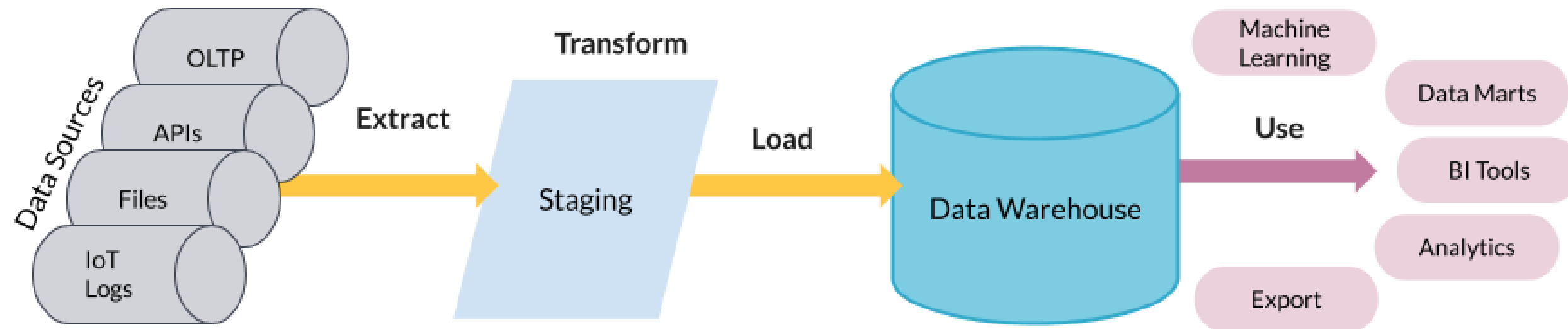


Data lakes

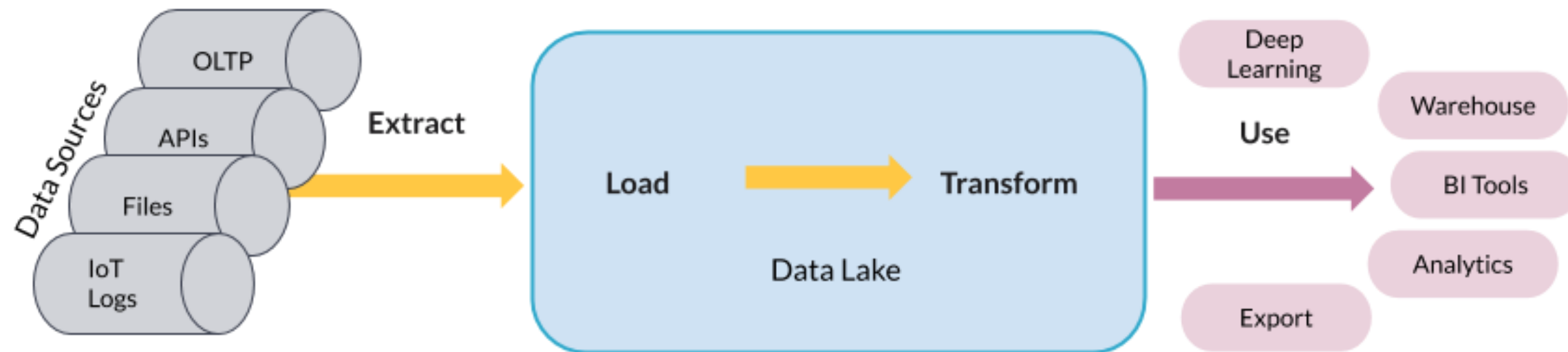
- Store *all* types of data at *a lower cost*:
 - e.g., raw, operational databases, IoT device logs, real-time, relational and non-relational
- Retains all data and can take up petabytes
- Schema-on-read as opposed to schema-on-write
- Need to catalog data otherwise becomes a **data swamp**
- Run **big data analytics** using services such as **Apache Spark** and **Hadoop**
 - Useful for deep learning and data discovery because activities require so much data



ETL



ELT



Let's practice!
DATABASE DESIGN

Database design

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What is database design?

- Determines how data is logically stored
 - How is data going to be read and updated?
- Uses **database models**: high-level specifications for database structure
 - Most popular: relational model
 - Some other options: NoSQL models, object-oriented model, network model
- Uses **schemas**: blueprint of the database
 - Defines tables, fields, relationships, indexes, and views
 - When inserting data in relational databases, schemas must be respected

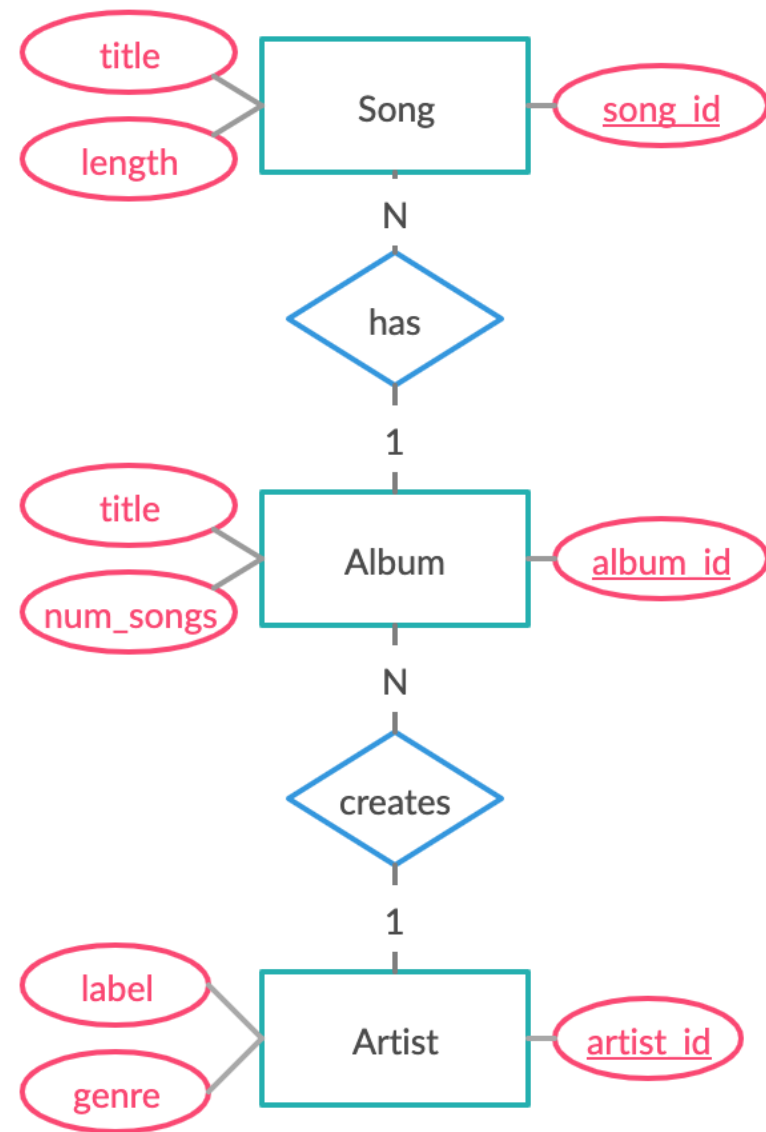
Data modeling

Process of creating a *data model* for the data to be stored

1. **Conceptual data model:** describes entities, relationships, and attributes
 - *Tools:* data structure diagrams, e.g., entity-relational diagrams and UML diagrams
2. **Logical data model:** defines tables, columns, relationships
 - *Tools:* database models and schemas, e.g., relational model and star schema
3. **Physical data model:** describes physical storage
 - *Tools:* partitions, CPUs, indexes, backup systems and tablespaces

¹ https://en.wikipedia.org/wiki/Data_model

Conceptual - ER diagram



Entities, relationships, and attributes

Logical - schema

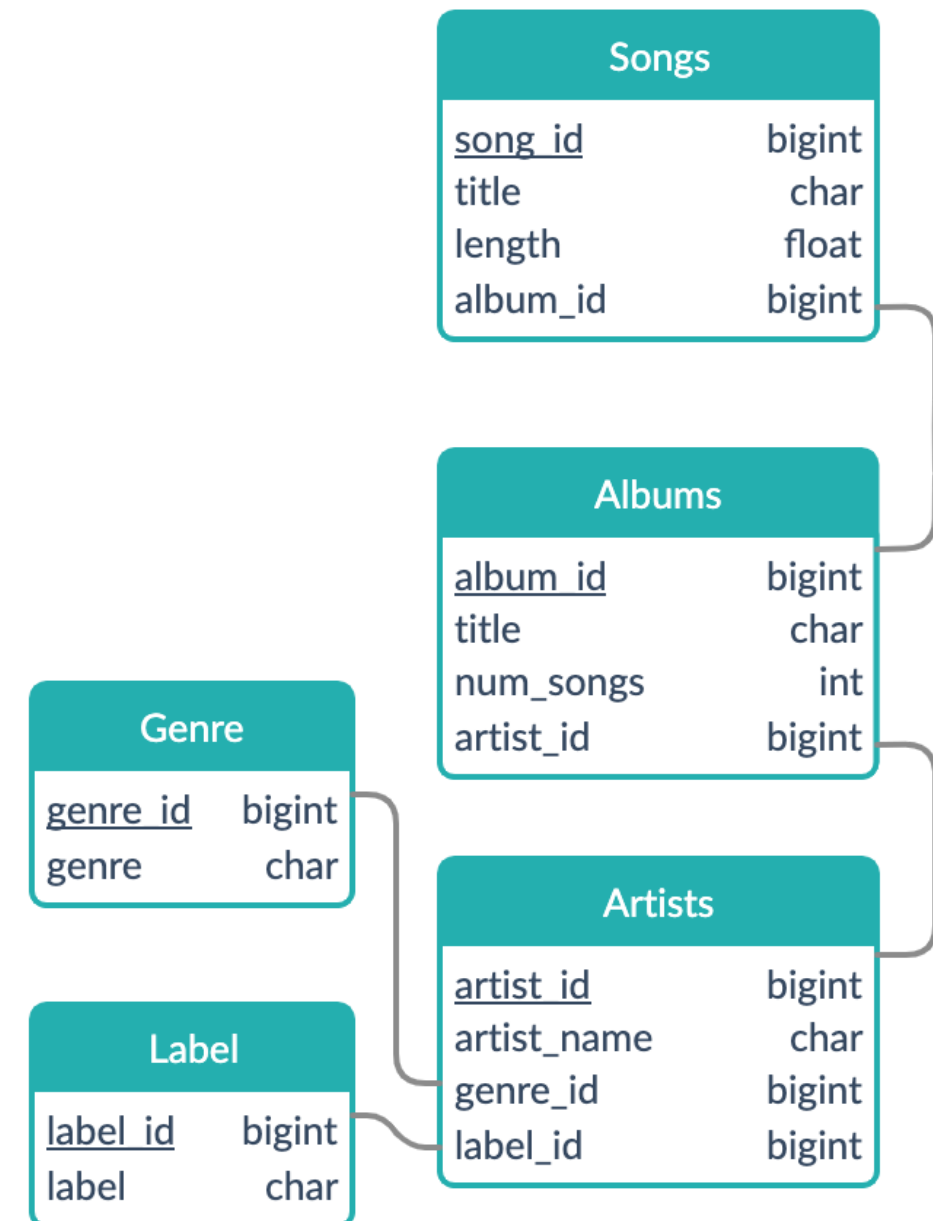


Fastest conversion: entities become the tables

Other database design options

Songs	
<u>song_id</u>	bigint
song_title	char
length	float
album_title	bigint
num_songs_album	int
artist_name	char
genre	char
label	char

Determining tables



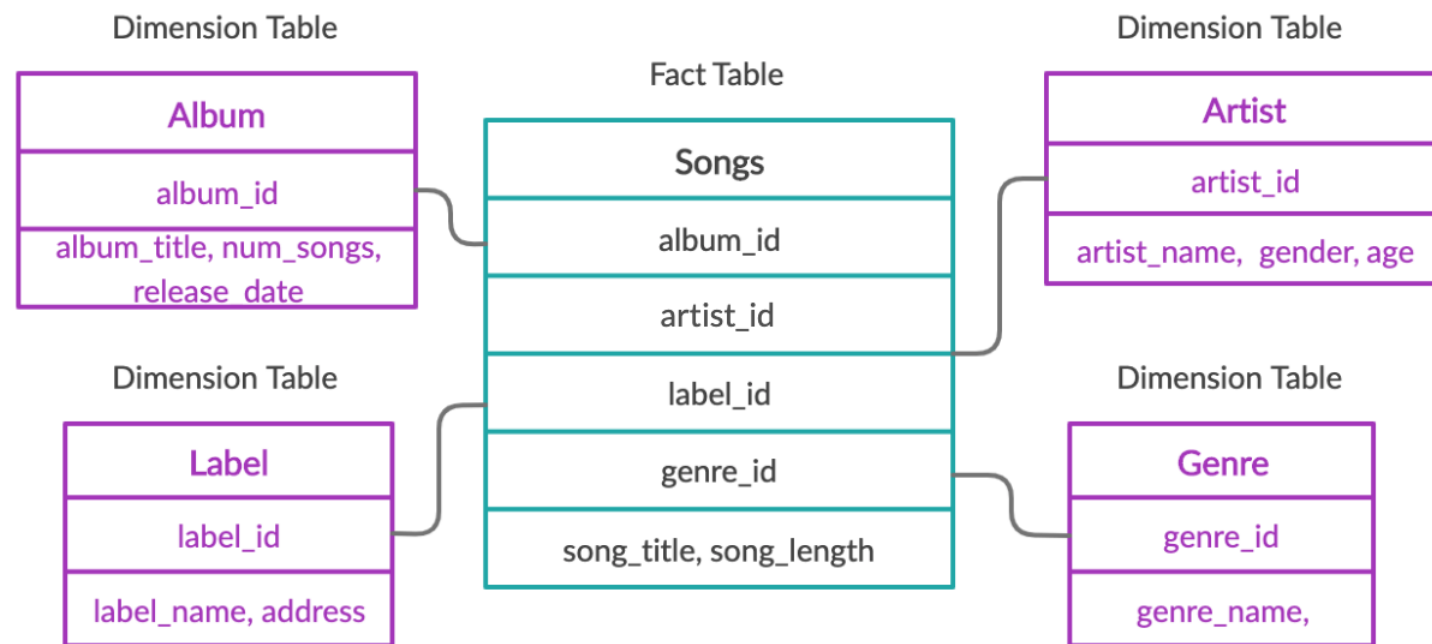
Beyond the relational model

Dimensional modeling

Adaptation of the relational model for data warehouse design

- Optimized for **OLAP** queries: aggregate data, not updating (OLTP)
- Built using the star schema
- Easy to interpret and extend schema

Elements of dimensional modeling



Organize by:

- What is being analyzed?
- How often do entities change?

Fact tables

- Decided by business use-case
- Holds records of a metric
- Changes regularly
- Connects to dimensions via foreign keys

Dimension tables

- Holds descriptions of attributes
- Does not change as often

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